CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2013 series

5014 ENVIRONMENTAL MANAGEMENT

5014/12 Paper 1, maximum raw mark 120

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

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General notes

Symbols used in Environmental Management mark schemes.

/ separates alternatives for a marking point – other valid ways of expressing the same idea are also credited

; separates points for the award of a mark

[3] indicates the number of marks available

italic indicates that this is information about the marking points and is not required to gain credit
italic text is also used for comments about alternatives that should be accepted, ignored or rejected

ora or reverse argument – shows that an argument from an alternative viewpoint will be credited

AW alternative wording, sometimes called 'or words to that effect' –
AW is used when there are many different ways of expressing the same idea

( ) the word / phrase in brackets is not required to gain marks but sets the context of the response for credit
e.g. (nuclear) waste – nuclear is not needed but if it was described as a domestic waste then no mark is awarded

<u>volcanic</u> underlined words – the answer must contain exactly this word

ecf error carried forward – if an incorrect answer is given to part of a question, and this answer is subsequently used by a candidate in later parts of the question, this indicates that the candidate's incorrect answer will be used as a starting point for marking the later parts of the question

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Section A

1 (a) (i) scrap aluminium (cans, pans etc.);

energy / electricity; **[2]**

(ii) only 5 % of the energy needed / uses less energy; **[1]**

(b) clearance of land for factories destroys habitats;
gas emissions pollute the atmosphere;
liquid wastes pollute water courses;
solid waste dumps can contaminate water supplies;
piles of waste / chimneys / large factory building look unsightly / are eyesores;

Mention with brief description = 1 mark
Fuller description / use of an example = 2nd mark
Max of 2 marks per way (except in exceptional cases)
Any combination of 1 and 2 mark answers to a max. of 4 marks **[4]**

(c) regulations and anti-pollution laws;
monitoring by government officials / agencies;
persecution of offenders with fines / prison terms;
closing down persistent offenders;
government support for renewable / alternative energy sources;

Mention with brief description = 1 mark
Fuller description / use of an example = 2nd mark
Any combination of 1 and 2 marks answers to a max. of 3 marks **[3]**

[Total: 10]

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2 (a) (i) accurate plots of 26.0 and 50.4; **[1]**

(ii) absence of vegetation means;
nothing to intercept the rain;
nothing to slow movement of rainwater downslope;
no plant roots to take up and use rainwater;
bare surface may be baked hard / compacted allowing little rain infiltration;

Any two **[2]**

(iii) although the ground vegetation cover in 2 might be denser, the grass is shorter, and has lower water needs than the trees in 4; also grass leaves are narrower and intercept less rain;

Some understanding = 1 mark
Clear explanation including direct reference to the grass = 2 marks **[2]**

(iv) amount of ground cover varies from none (before and immediately after planting) and most runoff, to complete cover (before harvesting) and low amount of runoff.

Clearly understood = 1 mark **[1]**

(b) best choice is land use 4 (bare fallow); soil conservation measures include;
mixed cropping / crop rotation;
planting trees in the context of acting as windbreaks tree crops (but not just tree planting);
addition of organic matter to maintain soil structure;
possibly contour ploughing and terracing since a slight slope is indicated;
mulching / covering soil with dead plant matter;

Name of method with a little description = 1 mark
Further description about reducing rates of soil erosion = 2nd mark
Any combination of 1 and 2 mark answers up to a maximum of 4 marks

Any other land uses choices – mark answer on its merits. In many cases the maximum mark possible is likely to be only 2, or at best 3 marks. **[4]**

[Total: 10]

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3 (a) (i) all areas are centred around the Equator;
never extends outside the tropics;
further description of location in terms of extent for individual areas;

any two valid descriptive points **[2]**

(ii) hot and wet all year;
further details about either or both of these; **[2]**

(b) (i) *for an increase in global warming:*
burning tropical forests releases carbon dioxide into the atmosphere;
carbon dioxide is a greenhouse gas;
traps in the atmosphere heat radiation from the Earth's surface;
the natural tropical rainforest is a huge carbon store / a larger carbon store than a palm oil plantation;

against an increase:
both forests and oil palm plantations are carbon neutral;

Three reasons along the lines outlined above. **[3]**

(ii) rainforest about 3 times as much / 315–320 tonnes compared with 100; **[1]**

(iii) *WWF member* – loss of biodiversity / habitats / species;

forest dweller – fear of being displaced elsewhere / loss of forest for food and many other needs; **[2]**

[Total: 10]

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4 **(a)** mainly grass with trees / shrubs in background;
further description of relative size of areas covered by each of these;
grass looks brown / dry;
sparse because bare ground / soil is showing in places;
umbrella shaped tree / acacia in centre at back of photo;
some shrubs / bushes look to be without leaves;

Three photo based descriptive points such as these.
3 × 1 mark **[3]**

(b) *population* – total of all the animals of the same species;

habitat – home of the animals; here the grassy savanna;

consumer – animals eating the grass / vegetation;

ecosystem – plants / animals interacting with the natural environment;
4 × 1 mark **[4]**

(c) animals damage vegetation by trampling, eating young shoots and stopping bushes, shrubs and trees from growing to full size;

some animals such as elephants strip bark off trees / pull down bushes and trees;

herds of animals trample on and compact bare soil, especially in the dry season, increasing soil erosion and weakening the soil structure, perhaps leading to desertification; also overgrazing;

vegetation loss can have further effects on the rest of the food chain;

animal wastes fertilise the soil; likewise when they die and decompose, animals help in seed dispersal;

1 mark for clearly stated reason
2nd mark for more detail about the reason
Any combination of 1 and 2 mark answers to a maximum of 3 marks **[3]**

[Total: 10]

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Section B

5 (a) (i) 1 North America 2 Oceania 3 Europe 4 Latin America 5 Asia 6 Africa; **[1]**

(ii) $26,210; **[1]**

(iii) values plotted on map matching the bands of shading;
key completed which matches the colours / shading used on the map;
quality, appearance and value of shading and completed map; **[3]**

(iv) 1 *Good fit* – between North and Latin America is best because two shading zones different; good also between Europe and Africa with one shading zone difference (and Africa has the lowest value so big difference in actual numbers). Likewise need to separate Oceania from Asia.

2 *Less good fit* – Asia is the obvious example because its average value is low, but some parts such as Russia and Japan are in the developed North. Line splits the continent in half.

3 *Comment about overall fit* – many variations possible. Good fit in separating out the developed continents of North America and Europe from the developing continents of Africa and Latin America i.e in western half of the map. Line less valuable in the east of the map where its course is forced to change from running east-west to north-south to accommodate the high development of Japan and Australia, to some extent making the North-South world label less valid.

Basically – 1 mark for each of parts 1, 2 and 3
4th mark from amount and quality of comment in part 3

It is possible, however, that the 4th mark can be distributed to 1 and 2 as well for a particularly precise answer. **[4]**

(b) (i) Average income per head; **[1]**

(ii) Best answer;
much more likely to suffer from water-borne diseases such as diarrhoea, typhoid and cholera (*i.e. 1 mark for recognising the importance of water-borne diseases*), since these spread as a result of drinking contaminated water, also caught by washing food / pots and pans / hands and face in dirty water.

Up to the full 3 marks

Related points provided that water-borne recognised; these diseases easily spread to other people in areas with poor sanitation, young children are most at risk with lower immune systems, disease reduces work ability of parents / increases poverty with less chance of affording medical attention.

Maximum 1 mark for these if water-borne not established; otherwise 2 marks possible.

References to water-based diseases such as bilharzia can also be made relevant by stressing that it is most likely to be caught by people working in polluted waters. Others such as Guinea worm are caught by people drinking contaminated water.

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Up to the full 3 marks, although the marks are harder to earn because the focus must be kept on access to clean (safe) water.

Water-bred diseases such as malaria are not relevant to this particular question.

3 × 1 mark for points made, bearing in mind restrictions stated above. **[3]**

(iii) Strongest support from values for China–Japan separation, particularly the huge difference between average income per head (over US$ 35,000), but also for birth rate, life expectancy and access to safe water the relative differences are large;

Average income per head (especially) would suggest that South Korea should now be placed north of the line, that it no longer belongs in the South;

Russia's position in the North can be questioned – life expectancy is well below that of all the others, even China, despite the very favourable people per doctor ratio;

How well? Individual candidates can hold different overall views, which are capable of being justified by the data. Perhaps the best view is that the line has its uses for splitting the world up into a simple two-fold division, but that at the country level there are always likely to be exceptions;

Quality of explanation based on study of measures in table = 1, 2 or 3 marks
Attempt with support to answer 'How well do the differences … ?' = 1 or 2 marks **[4]**

(c) (i) accurate plot of pie graph;
key completed for shading / colours used in pie graph; **[2]**

(ii) Developing countries are only receiving 10% of final price / only a small amount – what the coffee farmer in the tropics receives is the only source of income for the developing country;
Or
90% of the final price goes to companies in developed countries – for transport, manufacture and retail;

Understood = 1 mark **[1]**

(iii) Guaranteed price to the farmers for their crops, even when world market price falls – farmers know the minimum price they are going to get for their crops, which allows to survive when world market price falls; No longer at the mercy of world market prices over which the farmers have no control, and which change wildly;
Or
Helping them to find a market for their produce by putting them in touch with overseas importing companies from small producers; opening up a market for their produce, again safeguarding their income;

1 mark for describing the way from the source
2nd mark for explaining the advantage of Fair Trade **[2]**

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(iv) Ways referred to include community projects / outside toilets leading to improved sanitation / community canteens.
Explanation depends on choice.
For example, improved sanitation reduces risk of diseases spreading, making people ill and reducing their work productivity. Community canteens and a proper meal improves rural health thereby allowing villagers to devote more time to productive activities; children more likely to grow up healthier with the attendant social and economic benefits.

1 mark for naming supported by only outline explanation
2 marks for fuller explanation related to the named way. **[2]**

(v) would need to deal with too many different producers to obtain big amounts needed (compared with dealing with one large plantation owner);
small producers less able to guarantee reliable quality and consistent quantity;
more interested in profits than in helping local communities and paying farmers more;

Realises what the question is asking and/or identifies the most relevant comment = 1 mark
Understands the question well and make one or more valid suggestions = 2 marks **[2]**

(d) (i) A – food aid
B – non-governmental aid (NGO)
C – governmental (bilateral) aid
1 mark

(ii) A – short-term
B – long-term
C – long-term
1 mark

(iii) A – unsustainable
B – sustainable
C – allow either answer; instead mark this answer in part (iv)
1 mark **[3]**

(iv) *If unsustainable* – environmental and ecosystem damage for their creation, people relocated not always well looked for housing and good farmland, are of some relevance; more important are sediments build up in lake behind dam, its effectiveness is reduced with time, rivers below dam deprived of water and silt with further environmental consequences downstream, water often used many kilometres away in channel irrigation, sometimes leading to soil salinisation.

If sustainable – large dam will provide large water supplies / energy for many years, giving guaranteed water supplies in areas with dry or unreliable climates, allowing reclamation of desert land for food cultivation, assuring food and clean water supplies, encouraging economic development benefiting future generations.

Valid explanation for viewpoint chosen – 1 or 2 marks according to number of points made, or for one point well elaborated. **[2]**

(v) *Advantages*
alleviates immediate suffering after a natural disaster;
can be a life saver for poor people badly affected;

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well targeted development aid allows improvements that poor villages people could otherwise not afford;
examples of these such as deep wells, small dams, clinics, schools;
improve infrastructure for longer term economic development;

Points made such as these; 2 × 1 mark

Disadvantages
fosters dependence on outside help which cannot last for ever;
big money can be spent on large projects which benefit only a few people;
some 'tied aid' does not target the real needs of people or countries;
aid does not always reach remote rural communities where it is most needed;

Points made such as these; 2 × 1 mark

Allow 3:1 or 1:3 ratios provided that either answer includes reference to a specific example of a project, or named countries and relevant aid. **[4]**

(e) (i) Award 1 mark if the name of a country, or area or resort in a developing country is sufficiently precise. Country names such as India or Brazil would not be enough, but holiday islands such as the Maldives, Mauritius and Barbados would be acceptable.
Otherwise hold back the mark at this point; take the name into account when marking the second part.

(ii) Tourist attractions, either natural or human, and related to the named area
– up to 2 further marks with 1 mark already awarded in **(i)**
– up to 3 marks if definitely related to the area named in **(i)***, even though that was insufficiently precise for the mark in* **(i)**

How managed for a sustainable future?
Controls on destruction of the natural environment, such as National Parks, Forest and Biosphere reserves; management and maintenance; controlling numbers of tourists; training guides; building eco-lodges; ecotourism initiatives
– up to 3 marks if definitely related to the area named in **(i)**.

Without a valid location named in **(i)***, the maximum mark for an answer of a general nature is 2 marks, either 1 + 1 mark for attractions and management, or more likely up to 2 marks for management methods used in tourism.* **[5]**

[Total: 40]

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6 (a) (i) *climatic* – cyclone, drought and flood
tectonic – earthquake and volcano **[1]**

(ii) *cyclone* – tropical storm, with very low pressure, associated with high winds and heavy rainfall
earthquake – shaking of the ground surface; Richter scale measures the strength of the shaking
flood – large amount of surface water covering areas which are usually dry; areas under water after heavy rains
volcano – cone shaped mountain built up by eruptions of magma from the interior of the Earth.

4 definitions × 1 mark
5th mark for high levels of precision, or for the inclusion one or two particularly strong definitions. **[5]**

(iii) Possible choices are all of them, except earthquakes.

Examples;
cyclone – done by improved weather forecasting; use of weather satellites to view cloud patterns and track the course of the storm; possible to give advance notice and estimate chance of land area being hit. There is time for people to be evacuated to shelters or to move inland. Areas likely to be affected by cyclones are well known.

volcano – set up instruments to measure and monitor temperatures in and around the crater; use seismographs to register earthquake shocks prior to an eruption; look for visible signs of impending activity such as increased amounts of steam and gases coming from the crater.

No mark for the choice; all three marks describing how prediction can be done.
3 points along the lines of those suggested above; 3 × 1 mark **[3]**

(iv) The best choice is earthquake – while zones at high risk of earthquakes are well known since they coincide with plate boundaries, no one has been able to predict accurately when an earthquake will occur. The big shock is often the first;
the after-shocks are less powerful than the main earthquake, so there are usually no small warning shakes. Constant plate movement; impossible to know when the tension will be released

1 mark for choice of earthquake
2nd mark for an explanation

For the others, no mark for choice; 2 marks can be awarded for quality of explanation for any of the others. The choice will control the likelihood of enough explanation for both marks.
For example, for volcanoes past performance in eruptions may be no guide to the future; some do just erupt without notice and much history of doing so. Cyclones can change course at the last minute, or strengthen suddenly threatening areas that thought they were safe.

drought needs a longer time frame in which to occur. The annual rains might be late in arriving, but meteorologists cannot predict whether it will rain weeks or months ahead, later in the wet season.

Up to 2 marks for quality of the explanation. **[2]**

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(b) (i) correct plots of 90% and 98% on the bar graphs;
key completed for developed and developing countries which matches what is shown in the graphs; **[2]**

(ii) build shelters – cyclone / also tsunami;
train emergency rescue teams – earthquake;
store of staple foodstuffs – drought;
3 × 1 mark **[3]**

(iii) Possible reasons;
Poverty – lack of money so that governments are unable to afford to undertake the advance precautions, nor do they have the resources to deal with the immediate aftermath during which lives could be saved. Only the rich can afford to do things for themselves such as stronger houses.

Poverty cycle – people have been unable to recover from previous natural disasters, most noticeable in droughts; rainfall may have been below average for several years not giving a chance for food stocks to be built up.

Government attitude – many other problems to deal with in developing countries;
hazard protection is a lower priority than in developed countries where the social role of governments is greater.
Some governments too corrupt to help people.

Infrastructure and organisation – places affected are often more difficult to reach in developing countries making emergency aid arrival slower.

Geographical location – some of the natural hazards, noticeably tropical cyclones, affect latitudes where developing countries are more numerous than developed countries; droughts are made worse in the tropics by high rates of evaporation

High rates of population increase – pressure on the land and family food supplies exist even without natural hazards. Over-use of farm land, or shanty settlements on steep slopes, make the effects of some natural hazards worse.

Cumulative effects of repeated / different natural hazards – little time for recovery

The most widespread choice is likely to be poverty – maximum of 3 marks,
unless two very separate aspects of poverty are included as suggested above.
Otherwise – for each reason;
reason identified with outline explanation = 1 mark
more detailed explanation (could be helped by reference to an example) = 2 marks

Any combination of 1 and 2 marks answers up to the mark of 4. **[4]**

(iv) some hazards give economic benefits such as volcanoes for fertile soils, minerals, tourism, and are naturally attractive places to live;
people own land and property in the high risk areas; they cannot give up what may be their only assets; massive investments in big urban areas, too big to abandon;
attached to the area; have learnt to live with the risk; some can afford precautions;
belief that it won't happen to them; hazard only happens from time to time, and may be not at all where they live;

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Reasons / explanation – credit elaboration and use of examples.
Answer could refer to just one natural hazard with full elaboration, which covers different reasons.
3 valid points made: 3 × 1 mark **[3]**

(c) (i) Big size of flood – nothing like it since the 1920s, almost 100 years ago. **[1]**

(ii) North- south flood zone around the River Indus, extending the full length of the country from the northern boundary to the sea, strip of flooded land is 200km or more wide, only a few areas west and east of the Indus in the north and centre not affected, largest unaffected area in the south west / to the north and west of Karachi. Estimate of area of country affected such as about one quarter.

Points such as these taken from study of the map. 3 × 1 mark **[3]**

(iii) Rains in the mountains of the north were so great that the River Indus was still in flood 1000km further south i.e. related to the size of the monsoon rains in 2010. **[1]**

(iv) People in the centre and south had plenty of time to move away from the high flood risk areas on the sides of the rivers – it took time for the flood waters to be carried south by the River Indus. Only people in the mountain valleys of the north such as the Swat Valley had little warning.

Floods were on such a scale that nobody elsewhere in Pakistan could be unaware of them, no matter how poor communications might be in rural areas.

River valleys of large rivers such as that of the Indus are very favourable for farming and settlement – that is why so many people were affected. Most people in developing countries like Pakistan are farmers.

Two different reasons such as these or one reason well elaborated.
2 × 1 mark **[2]**

(d) (i) *Short-term effect on cropland – one from*:
loss of crops; farmland covered by mud / stones / tree trunks; nothing to sell

Long-term effect on cropland – one from:
layers of fertile silt deposited, underground water stores filled up with waters making it more abundant for future crop use

1 + 1 mark **[2]**

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(ii) Key part of the report refers to the rural poverty trap; 'poor people with little before the flood were left with nothing after it' ... Also 'Pakistani poor were the worst hit' These, along with all the further information about losses to crops, animals, belongings and infrastructure, suggest little chance of getting out of the poverty cycle. This is reinforced by references in the report to lack of government help and aid from outside.

Great challenges in clearing the farmland of mud, stones, tree trunks and other debris for poor people without anything.

Only glimmers of hope in the report are the renewal of soil fertility from the silt deposits, plus more plentiful ground water supplies for irrigation in future years.

Choice and use of content from the report related to the question's poverty cycle theme, but with limited added explanatory comment – 1 or 2 marks

Good choice of content from the report, with the candidate's view about the chances of getting out of the poverty cycle supported and enhanced by comment – 3 or 4 marks **[4]**

(e) Earthquakes – little danger from shaking of the ground in open spaces; plenty of these in rural areas, but in short supply in cities. Falling debris and collapsing buildings are what do the damage and kill and injure people in earthquakes.

Droughts – most significant for people who rely most upon water supplies from rainfall and rivers; they are mainly farmers needing water for crops and pastures for livestock. In urban areas, domestic water use is much lower and water is often piped in from mountains (where drought is less likely) or from man-made stores such as reservoirs.

Possible for candidates to use examples of other natural hazards in their answers.
Floods – the previous example of floods in Pakistan showed the serious effects for farmers and rural communities
Volcanoes – effects often worse in rural areas, closer to the the volcano, and farmland can be covered by lava and ash; eventually fertile, but it takes time to break down into soils that can be used.
Cyclones – hazard for which effects on urban and rural areas may be equally severe;
coastal areas that take the direct hit can be devastated whether urban or rural.

Answer may be more about the different effects of natural hazards, only weakly related to the theme of differences between urban and rural areas. Or it may be all about just one natural hazard such as earthquakes. Overall limited coverage.
1 or 2 marks

Answer better focused on the theme of variations between rural and urban areas, supported by references to at least two natural hazards and their differing effects.
3 or 4 marks **[4]**

[Total: 40]

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